Argumentation
https://doi.org/10.1007/s10503-018-9464-z

The Thesis of the Effectiveness of Quasi-logical Arguments

Iva Svačinová[1]

© Springer Nature B.V. 2018

Abstract
The article focuses on the new rhetoric category of quasi-logical arguments, defined as arguments similar to logical or mathematical demonstrations, and therefore having an effect on the audience. Connecting the *similarity* of arguments to formal demonstrations with the claim of *effect* on audience is conceived in this article as the *thesis of effectiveness of quasi-logical arguments*. The components of the thesis are reconstructed and analyzed, and their precise definitions are proposed. The analysis shows that the category of quasi-logical arguments is ambiguously defined and currently includes three different concepts based on a different understanding of similarity. Subsequently, it is showed that based on Perelman and Olbrechts-Tyteca's works, the explanation of quasi-logical arguments' effect on an audience can be offered in terms of socially shared standards of community or in terms of individual processing of arguments. It is shown that these explanations can be in conflict each other regarding the assumption of the addressee's acquaintance with formal principles.

Keywords Argument schemes · Effectiveness · New rhetoric · Quasi-logical arguments

1 Introduction

Quasi-logical arguments are a category of arguments defined in the new rhetoric, the model of argumentation submitted by Chaïm Perelman and Lucie Olbrechts-Tyteca. They are conceived as arguments that are (a) similar to formal demonstrations and thus (b) having an effect on the audience. Although there is an evident interest in the new rhetoric's typology of schemes and many of them were taken and used in modern theoretical approaches to argumentation (cf. Walton et al. 2008; Kienpointner 1993, 2002; Warnick and Kline 1992), the category of quasi-logical arguments

✉ Iva Svačinová
iva.svacinova@gmail.com

[1] Department of Philosophy and Social Sciences, Philosophical Faculty, University of Hradec Králové, Rokitanského 62, Hradec Králové, Czech Republic

Published online: 17 July 2018

is paid little attention by modern theorists of argumentation. This paper deals with why this is the case. The connection between the similarity to formal demonstrations and the claim of its effectiveness is conceived in the text as the *thesis of effectiveness*. The components of the thesis are reconstructed and analyzed, and their precise definitions are proposed. The answer is offered that (a) the quasi-logical argument category is ambiguously defined and currently includes three different concepts of similarity; (b) the explanation for the effect of quasi-logical arguments is vague too and Perelman and Olbrechts-Tyteca's comments enable us to see it from two different perspectives.

The paper is as follows. Section 2 presents the historical impulses influencing the new rhetoric and quasi-logical arguments' position in this theoretical framework. In Sect. 3, a preliminary version of the thesis of effectiveness of quasi-logical arguments is formulated, based on Perelman and Olbrechts-Tyteca's definitions of quasi-logical arguments.

Sections 4 and 5 are devoted to the reconstruction and analysis of the thesis of effectiveness' components: similarity and effectiveness. In Sect. 4 it is pointed out that the similarity between arguments and formal demonstrations is reflected in different ways. Different concepts of similarity can be identified with respect to the addressee, who discerns the quasi-logical scheme. Three concepts of quasi-logical argument are distinguished: theoretical, practical, and interpreted (Sects. 4.1–4.4).

In Sect. 5, the claim of the quasi-logical scheme's effectiveness is reconstructed and the specified definition of the scheme's effectiveness is offered. The effectiveness is conceived as audience's acknowledgment of relevance of the connection of the elements in an argument. Subsequently, two possible explanations for the source of acknowledging the quasi-logical elements' connection in argument are investigated, based on the emphasized aspect of receiving an argument by the audience, in terms of socially shared standards of community or in terms of individual processing of arguments (Sects. 5.1–5.1.3).

2 Programmatic Distinction of Logic and Argumentation in the New Rhetoric

The new rhetoric[1] from the Belgian authors Chaïm Perelman and Lucie Olbrechts-Tyteca is considered to be one of the first modern approaches to exploring argumentation (for an overview see van Eemeren 2009). The authors developed the approach in the post-war years and published their results after a decade of development in 1958, in the extensive volume of *The New Rhetoric*.[2] The key motif shaping the

[1] The term "the new rhetoric" is used here in reference to Perelman and Olbrechts-Tyteca's theoretical approach to argumentation, the term "*The New Rhetoric*" refers to their seminal book (1969).

[2] The publication of the book was preceded by several studies in which the new rhetoric's approach to argumentation was formed (cf. Perelman 1954, 1955; Perelman and Olbrechts-Tyteca 1951, 1957). In 1958, the book was published in French as *Traité de l'argumentation: La nouvelle rhétorique*, in 1966 it was translated into Italian. However, it gained its publicity with its translation into English in 1969.

approach of the new rhetoric, which is most clearly reflected in the concept of quasi-logical arguments, is a sharp distinction between logic and argumentation.

We can identify the reasons for the distinction historically. The stimulus determining the formation of the new rhetoric was the philosophical context of the 1940s and 1950s dominated by logical empiricism. Perelman and Olbrechts-Tyteca blame the contemporary logicians and philosophers for a narrow understanding of reasonableness, which is nowadays referred to as the *geometric* concept of reasonableness: an argument is considered valid if it is logically valid, i.e., if its conclusion can be derived from the premises (cf. Toulmin 1976, p. vi; van Eemeren and Grootendorst 1988, p. 275). The authors find the origins of such a narrow understanding of reasonableness in the work of Rene Descartes, who understands the derivation of truth from self-evident truths to be the ideal of rationality (cf. Perelman and Olbrechts-Tyteca 1969, p. 1). Such a narrow concept is, according to these authors, characteristic of the Western philosophy in the last three centuries. Its direct inheritor can be the concept of a formal, logical demonstration in which reasoning is understood as a sequence of propositions in which each member of the sequence is either one of the axioms of the system or results from the preceding members of the sequence by derivation rules.

Perelman and Olbrechts-Tyteca consider such a narrow concept of reasonableness as "a perfectly unjustified and unwarranted limitation of the domain of action of our faculty of reasoning and proving." (1969, p. 3) According to them, it leads to an undesirable consequence: activities like submitting the reasons in favor of values or implementing actions would not be considered as reasonable activities. However, according to the authors "it would scarcely occur to anyone to deny that the power of deliberation and argumentation is a distinctive sign of a reasonable being (…)." (Perelman and Olbrechts-Tyteca 1969, p. 1).

The programmatic starting point for formulating the new rhetoric is extending the concept of reasonableness to include the activities of practical reasoning that do not need to fulfill the demands of formal demonstrations. To achieve this goal, Perelman and Olbrechts-Tyteca propose to distinguish between two different types of reasonable activities. In their approach, they follow Aristotle's distinction between analytic reasoning developed in *Analytics*, and the dialectical reasoning developed in his *Topics* and *Rhetoric* (cf. Perelman 1979, p. 9; Perelman and Olbrechts-Tyteca 1969, p. 5). Aristotle's thesis, according to which, analytic reasoning (proof) and dialectical reasoning (argumentation) are two different methods of reasoning, is updated in the new rhetoric with respect to the contemporary context. Perelman and Olbrechts-Tyteca identify *formal logical demonstration* and *practical reasoning* as two different ways of reasoning (cf. Perelman 1986, p. 7; Perelman and Olbrechts-Tyteca 1969, pp. 13–14).

A *demonstration* is considered by Perelman and Olbrechts-Tyteca as "processes by means of which the proposition can be obtained as the final expression of a deductive series, which had its first elements provided by the constructor of the axiomatic system within which the demonstration is accomplished." (1969, p. 14; cf. Perelman 1979, p. 10) If a mathematician or a logician wants to successfully demonstrate a proposition, she has to show that the proposition is provable in the axiomatic system. In order to do so, she only needs the initial elements of the system:

a predefined set of axioms, and a set of syntactic and derivation rules. Checking whether or not the proposition is provable in the axiomatic system can ideally be done mechanically (cf. Perelman and Olbrechts-Tyteca 1969, p. 13).

In order not to thwart this ideal, axioms and rules need to be defined in the most unambiguous way. Perelman and Olbrechts-Tyteca therefore link the demonstration's characteristics with the achievements of modern logic identified by Bochenski: the demonstration is implemented in an artificial language, is formalistic, i.e., its symbols are content-empty, and objectivistic, i.e., dealing only with objective characteristics such as truth, falsity, probability, necessity, independent of human attitudes and beliefs (cf. Bochenski 1981; Perelman 1989, p. 9).

Perelman and Olbrechts-Tyteca's idea of formal demonstrations is therefore relatively narrow: they are proofs in a formal language associated with an identifiable axiomatic system. As the ideal type, one can consider a series of logical/mathematical formulas that capture the derivation from the initial set of axioms to the proven proposition. Such a derivation process, in their view, can be judged to be correct/incorrect, given the initial set of rules and axioms. Since the domain of mathematics and logic is linked to the provability of propositions by the authors, it is obvious that other types of reasoning are not adequate for this goal or might even diminish it.

Practical reasoning or *argumentation* are defined by Perelman and Olbrechts-Tyteca regarding their purpose as "discursive techniques allowing us *to induce or to increase the mind's adherence to the theses presented for its assent*" (Perelman and Olbrechts-Tyteca 1969, p. 4, italics in original). This purpose is defined in contrast to the purpose of the demonstration:

> In argumentation, it is not a matter of showing (as it is in demonstration) that an objective quality (such as truth) moves from the premises toward the conclusion, but rather it is a matter of showing that one can convince others of the reasonable and acceptable character of a decision, based on what the audience already assumes and based on the theses to which it adheres with sufficient intensity. Persuasive discourse therefore aims at a transfer of adhesion, of a subjective quality which may vary from mind to mind. (Perelman 1989, p. 11)

The main difference in comparing with demonstration is therefore the claim that argumentation is essentially linked to the concept of an audience: it necessarily presupposes the existence of an addressee. Its goal is to have an effect on the audience, i.e., ideally to bring the audience to accept the standpoint ("thesis"). The second explicit definition is formulated by Perelman regarding the domain in which the argumentation occurs. Argumentation is defined as "the technique that we use in controversy when we are concerned with criticizing and justifying, objecting and refuting, of asking and giving reasons." (Perelman 1980, p. 108) For argumentation, therefore, the conflict of opinion or controversy is essential, i.e., the existence of an audience that has not accepted the standpoint yet.

Different general characteristics of the argumentation are linked with a different purpose and domain. Compared to demonstration, argumentation is characteristically formulated in natural language and the condition of unambiguity is not binding on it: in a communication, it is permissible to interpret one expression in different ways. At the same time, the authors point out that it is entirely relevant to count

on the audience's subjective interests. Argumentation is not a formalistic play with content-empty symbols. According to the authors, "when it is a question of arguing, of using discourse to influence the intensity of an audience's adherence to certain theses, it is no longer possible to neglect completely, as irrelevancies, the psychological and social conditions in the absence of which argumentation would be pointless and without result" (Perelman and Olbrechts-Tyteca 1969, p. 14). Perelman shows symmetrically that formal demonstrations are not an appropriate means in practical reasoning:

> It is just as foolish to ask a rhetor who wishes to persuade to utilize mathematical demonstrations in his speech as it is for a mathematician to substitute discourse for demonstration. It follows, therefore, that there are two different domains of reasoning: (1) the domain of the immutable and the unchangeable as epitomized by mathematics, formal logic, and metaphysics; and (2) the domain of practical reasoning where it is necessary, as in the case of politics, to decide, to choose, or to deliberate with others or ourselves. Here only practical reasoning is available. (Perelman 1986, p. 7)[3]

While for logical demonstrations, the geometric concept of reasonableness is adequate because it is developed with respect to them and therefore it seems entirely relevant to assess logical demonstrations with respect to the standards of logical derivation, for argumentation which is defined in contrast to logical demonstration, it is not the case. In this way, the new rhetoric programmatically justifies the attempt to "complete" the logic by developing a model for argumentation that focuses on a different type of reasonableness. A theory of argumentation should focus on those phenomena of non-analytic thinking that aim to persuade or convince the audience (cf. Perelman and Olbrechts-Tyteca 1969, p. 4).[4]

The geometric concept is replaced in the new rhetoric by the concept which is nowadays called the *anthropological* concept of reasonableness which "takes the rationality conception of a specific community as its starting point and bases its judgments on empirical evidence. Argumentation will then be regarded as valid if its point of departure and presentational layout agree with the standards of reasoning consented by the community concerned." (van Eemeren et al. 1996, p. 23) The transition to the anthropological concept of reasonableness has fundamental implications on the level of assessment for the validity of argumentation. While in the geometric concept the criteria of validity are given independently to the evaluator by the

[3] With regards to different standards of precision and exactness, it seems suitable to add that already Aristotle made similar remarks in his Nicomachean Ethics: ἡ μὲν οὖν μέθοδος τούτων ἐφίεται, πολιτική τις οὖσα. λέγοιτο δ' ἂν ἱκανῶς, εἰ κατὰ τὴν ὑποκειμένην ὕλην διασαφηθείη: τὸ γὰρ ἀκριβὲς οὐχ ὁμοίως ἐν ἅπασι τοῖς λόγοι ἐπιζητητέον, ὥσπερ οὐδ' ἐν τοῖς δημιουργουμένοις. This then being its aim, our investigation is in a sense the study of Politics. Now our treatment of this science will be adequate, if it achieves that amount of precision which belongs to its subject matter. The same exactness must not be expected in all departments of philosophy alike, any more than in all the products of the arts and crafts. (Aristotle, Nicomachean Ethics, Book 1, 1094b 10–14). I wish to thank my anonymous reviewer for this point.

[4] For theoretical elaboration of difference between logical and rhetorical perspective see Wenzel (1990, 1992).

formal system's set of rules, in the anthropological concept, the decision concerning validity is shifted to the addressee of the argumentation. The argument is valid if it is accepted by the audience. The assessment of whether or not the argument is valid is therefore in the new rhetoric entirely dependent on the existence of the audience and its character (cf. van Eemeren et al. 1996, p. 98).

3 Definition of Quasi-logical Arguments and the Thesis of Effectiveness

Rejecting the geometric concept of reasonableness for the domain of argumentation is based on the assumption that formal demonstrations are not used in practical justification's domain. According to Perelman and Olbrechts-Tyteca, formal demonstrations in the sense of logical formula sequences simply do not occur in practical reasoning because of the different purpose, properties, and relevant domains. Against this background, however, in the new rhetoric a thesis Perelman and Olbrechts-Tyteca advance that presenting arguments in a way reminding people of formal demonstrations, is a strategy that can contribute to the audience's acceptance of the presented standpoint (cf. Perelman and Olbrechts-Tyteca 1969, p. 193). The authors call the category of arguments resembling formal demonstrations *quasi-logical arguments*.

Quasi-logical arguments is at the most general level defined as an argument scheme, the *effectiveness* of which for the audience is explicitly related to the *similarity* of these arguments to formal demonstrations (cf. Perelman and Olbrechts-Tyteca 1969, p. 193). In the new rhetoric model, quasi-logical arguments fall under the category of the so-called associative schemes in which separate elements are brought together (cf. Perelman and Olbrechts-Tyteca 1969, p. 190). In quasi-logical arguments, this is implemented in a way similar to what we know from mathematical or logical operations: arguments are formulated in a way that the relations between their elements remind people of relations used in logical or mathematical demonstrations.[5]

The authors point out decisively that quasi-logical arguments are not identical to logical or mathematical demonstrations. They are fully in accordance with Perelman and Olbrechts-Tyteca's definitions of the goal, the characterization, and the domain of argumentation: they are addressed to the audience, formulated in natural language, and occur in domains connected with persuasion and values (cf. 1969, pp. 13–14; Perelman 1989). The prefix *quasi-* refers only to a certain similarity with formal demonstrations. The arguments are seemingly logical, but do not fulfill (and do not intend to fulfill) the characteristics or demands made in formal demonstrations.

[5] Haynes points out that rather than "quasi-logical arguments", it would be more appropriate to use the more general term "quasi-formal arguments", given that the category includes the arguments imitating logical and mathematical demonstrations. Perelman and Olbrechts-Tyteca's terminology, in which both quasi-logical and quasi-mathematical arguments fall under the general category of quasi-logical arguments, is somewhat confusing in this respect (cf. Haynes 1981, p. 221).

Perelman and Olbrechts-Tyteca associate the quasi-logical argument form with the claim of its effect on the audience. Quasi-logical arguments "lay claim to a certain power of conviction, in the degree that they claim to be similar to the formal reasoning of logic or mathematics" (Perelman and Olbrechts-Tyteca 1969, p. 193). According to the authors, "since there are formal proofs of recognized validity, quasi-logical arguments derive their persuasive strength from their similarity with these well-established modes of reasoning." (Perelman and Olbrechts-Tyteca 1969, p. 193) The claim of the relationship between similarity and effectiveness is in this text considered as the *thesis of the effectiveness of quasi-logical arguments*:

> [Thesis] Arguments that are *similar* to formal demonstrations are *effective* with respect to an audience, i.e., they contribute to the audience's acceptance of the thesis.

The definitions of quasi-logical arguments presented by Perelman and Olbrechts-Tyteca can therefore be divided into two consecutive definitions. They are:

(a) *similar* to formal demonstrations and
(b) *effective* with respect to an audience.

Let us consider them as two constitutive components of the thesis: *similarity* and *effectiveness*. From a theoretical point of view, components of the thesis are associated with separate issues and it seems relevant to deal with these components separately. The first component of the thesis identifies arguments in general, through their relationship to formal demonstrations: they are arguments similar to formal demonstrations. Therefore, it seems necessary to specify how we should conceive the similarity: in which aspects may the arguments be considered as similar to formal demonstrations?

From a theoretical point of view, understanding what a similarity can be is crucial at the level of *identifying* and *reconstructing* quasi-logical arguments. Our initial understanding of an argument's similarity with formal demonstrations influences the way we formulate the standards and clues to identify the given phenomena in data (included training of encoders) and the way we eventually reconstruct these phenomena.

The second component combines these characteristics with the potential effect on the audience, i.e., increasing the submitted standpoint's acceptability for the intended audience. However, to understand how quasi-logical arguments can affect the audience, it seems necessary to examine the general definition of the effectiveness of the arguments and pay attention to the reasons why Perelman and Olbrechts-Tyteca suppose that arguments resembling formal demonstrations can be an effective means of persuasion.

From theoretical point of view, the question of how a quasi-logical argument's effectiveness could be understood is crucial at the level of *analysis* and eventually *evaluation,* when the analysts try to explain why the investigated empirical argument was or was not accepted by the particular audience, under given conditions in the particular context.

4 An Argument's Similarity to Formal Demonstrations

How can we define the property of being 'similar to formal demonstration' in the new rhetoric? In Perelman and Olbrechts-Tyteca's view, the speaker can achieve an argument's similarity to a mathematical or logical demonstration by manipulating the argument in two ways. She can make the argument similar to the demonstration *lexically*, by homogenizing the used expressions and thus giving the impression of simplicity and unambiguity; and/or *structurally*, by assimilating the argument structure to a particular formal demonstration's structure, thus giving the impression that it is a particular formal demonstration being implemented (cf. 1969, p. 194).

By adopting an anthropological concept of reasonableness, Perelman and Olbrechts-Tyteca attribute a key role to the audience in assessing the arguments. Whether the arguments are considered similar to a formal demonstration or not depends on the distinctive and addressee's interpretative abilities. The concept of similarity as depending on the audience's receptive abilities occurs, however, in the new rhetoric in more than one variant. The criterion on which we can distinguish the variants is the question of *which* audience is assessing the similarity of the arguments. In order to answer this question, we can also identify various quasi-logical argument concepts.

In the following sections, three quasi-logical argument concepts are distinguished, based on different concepts of similarity relating to the audience. We can distinguish between the theoretical concept for which we can find support in the theoretical passages of *The New Rhetoric*, the practical concept on which the empirical part of *The New Rhetoric* and Perelman's *The realm of rhetoric* is based, and the interpreted concept that can be found in the interpreters and popularizers' work in the new rhetoric.

4.1 Theoretical Concept of Similarity

The theoretical concept of similarity can be found in the theoretical passages of *The New Rhetoric*,[6] in which the audience's concept is presented as "the ensemble of those whom the speaker wishes to influence by his argumentation." (Perelman and Olbrechts-Tyteca 1969, p. 19) According to the authors, for an addressee's persuasion or conviction and therefore a successful argument, we need to consider the beliefs of the addressee—her starting points ("points of departure") and schemes she recognizes as a relevant connection or separation[7] of elements (Perelman and Olbrechts-Tyteca 1969, pp. 23–24, 45). In the theoretical concept, an audience's plurality is introduced: different audiences can have not only different sums of starting

[6] It is interesting that in Perelman's other works presenting quasi-logical arguments, considerations dealing with the possibility of differing audience's scheme perception is rather missing (cf. Perelman 1982, Ch. 6, 7, 1979, pp. 19–21).

[7] The new rhetoric considers dissociations, in which the unified elements are also separated as argument scheme types (cf. Perelman and Olbrechts-Tyteca 1969, p. 190; van Rees 2008).

points but also a different perception of what scheme is implemented in the argument.[8] One argument can be perceived by two audiences as implementing different schemes:

> However, we are convinced that these same arguments could be analyzed differently, in accordance with other planes of cleavage. And this is because there is no reason why a single statement cannot be regarded as capable of expressing several schemes which would act at the same time on the minds of different persons – even on a single hearer. (Perelman and Olbrechts-Tyteca 1969, p. 188)[9]

The concept of discernment of the argument scheme as completely dependent on the audience's receptive abilities is repeated in §45, which is a general introduction defining the category of quasi-logical arguments in *The New Rhetoric*:

> It should be repeated in this connection that one and the same argument can be understood and analyzed differently by different hearers, and logical structures can be regarded as mathematical, and vice versa. (Perelman and Olbrechts-Tyteca 1969, p. 194)

Whether we can attribute a similarity with the logical/mathematical demonstration to an argument is therefore, according to the theoretical concept, completely dependent on the addressee's receptive abilities. The question of whether the argument is similar to a particular formal demonstration is, in this sense, translated into the question whether an audience discerns an argument as similar to a particular formal demonstration. The property of the argument 'to be quasi-logical' or, more specifically for example 'to implement the argument scheme of arguments of reciprocity' is therefore understood as an *audience-dependent* property: we can attribute it to the argument solely depending on whether the audience discerns it as such. The same argument may or may not have this property if its addressees are two different audiences.[10]

In the theoretical concept, we can define similarity with respect to the audience that is the addressee of the argumentation as follows:

[8] Perelman and Olbrechts-Tyteca distinguish between two types of audience: universal and particular. The universal audience is the "whole of mankind, or at least, of all normal, adult persons" (1969, p. 30), particular audiences are people or groups with specific sums of beliefs or value preferences: to persuade the universal audience, it is necessary to choose factual premises. A discourse centered on values alone can never reach a universal audience—concrete values can never be considered binding for all human beings in all circumstances (cf. Gross and Dearin 2003, p. 36). Perelman and Olbrechts-Tyteca's original criterion for distinguishing between universal and particular audiences is therefore the character of audience *starting points*, not the audience's ability to recognize the schemes.

[9] In another place, Perelman and Olbrechts-Tyteca conceive: "In most cases, (…), we are simultaneously aware of more than just one way of conceiving the structure of an argument" (1969, p. 187).

[10] In this sense, the property 'to be quasi-logical' can never be considered a property dependent on an argument's superficial features.

> [Similarity$_1$] The argument implements a quasi-logical argument scheme when a connection of elements in an argument is *discerned* by an audience as similar to a particular formal demonstration.

Let us add a consideration concerning the character of discernment. According Perelman and Olbrechts-Tyteca, the similarity in connecting the elements in an argument to a formal principle can be discerned *readily* by the audience if the similarity is deliberately accentuated by the speaker. It may, however, be *less noticeable* if the speaker does not emphasize the similarity:

> Quasi-logical arguments are presented in a more or less explicit manner. Sometimes the speaker will mention the formal reasoning to which he makes reference, availing himself of the prestige of logical thought; sometimes the formal reasoning provides only an underlying texture. (Perelman and Olbrechts-Tyteca 1969, p. 194)

Moreover, the scheme's character does not need to be reflected and distinguished precisely by the audience: the audience does not to have to know exactly the formal principle whose similarity is used in connecting the elements or that the way the elements are connected is similar to a particular formal principle. In the new rhetoric, the discerning the scheme can be on a different degree of conscious reflection. Perelman and Olbrechts-Tyteca admit that usually, addressees do not deal actively with investigating the character of the connection of the elements in an argument presented to them. According to the authors, "it is possible (…) that these schemes *are effective without being clearly perceived* and that only an attempt at clarification, which is rarely performed, would enable the speaker, and especially his hearers, to become aware of the mental schemes which they are using or which are acting upon them (…)." (Perelman and Olbrechts-Tyteca 1969: 188, italics added, I.S.)

Note that defining quasi-logical arguments by their perceived similarity to formal demonstrations is inevitably ambiguous: in what ways and to what extent does the argument need to be assimilated to a logical demonstration to be perceived (with a different level of conscious reflection) by an audience as similar to it? And in what ways does it not need to be? Among the informal arguments formulated in natural language and formal demonstrations usually formulated in the artificial language of logic or mathematics, there is a fundamental gap. It can be bridged through a specific set of interpretative choices that express the audience's "receptive grid". These interpretative choices capture the specific features in the formulation of an argument sufficient for an addressee to discern the argument as similar to the formal demonstration. In other words, an addressee should ideally (at least in case of conscious reflection of scheme) be able to offer a list of such properties of arguments whose presence leads her to discern the argument as similar to a formal demonstration.

However, such a list is entirely dependent on the audience's preferences: on the extent of the syntactic and semantic variability of arguments of which the audience is willing to perceive the arguments as similar to formal demonstrations. Preferences may vary from audience to audience. In other words, assuming the addressee is familiar with the logical and mathematical demonstrations, it still depends entirely on her preferences whether or not a certain argument—formulated in a

natural language—is regarded as similar to a given formal demonstration or not. Consider two types of preferences: (a) the permissible extent of an argument's syntactic variability with respect to syntax of imitated formal demonstrations and (b) the permissible range of semantic interpretations in the elements of imitated formal demonstrations.

(a) Similarity does not mean identity, as Perelman and Olbrechts-Tyteca emphasize (cf. 1969, p. 193). Therefore, the argument does not need to copy the syntax of formal demonstration completely. It is sufficient if the composition and the structural layout of the elements in the argument is, to a certain extent, approaching the syntax of formal demonstration. However, Perelman and Olbrechts-Tyteca do not specify which elements and how many of them need to be preserved in the structural layout of the argument to still be perceived as implementing a particular quasi-logical argument scheme. They move this decision to the audience. Potentially, arguments that implement various variations which more or less resemble the formal demonstration's syntax can fall under the category of quasi-logical arguments. However, the permissible extent of these variations is not generally definable and is based on the preferences and tolerance of the audiences. Let's illustrate this with the following examples[11]:

(1) Since we [Europeans] find the customs of the Persians strange, shouldn't they be surprised at ours?
(2) They require of themselves toward their inferiors the same feelings that they ask of their superior.

Because in both cases there is accentuated mutual relationship between some individuals, it can be said that both examples could potentially remind an audience of the logical relationship of symmetry. According to that principle, relationship R is symmetrical if and only if, whenever it is true that there is a relationship R between x and y, it is true that there is a relationship R between y and x. Formally: $Sym(R) =_{def} R(x, y) \rightarrow R(y, x)$. However, we can say that example (1) is more approaching the logical principle of symmetry than example (2).

We can analyze the structural layout between relations in the fragment (1) in the following way: x = Europeans, y = Persians; R = to be surprised at customs of other side. Then, we can say, that the relationship R between x and y is reciprocal and follows the layout $R(x, y) \rightarrow R(y, x)$[12]:

However, in fragment (2), it seems accurate to analyze it capturing the relationship between three different actors: x = they, y = inferiors, z = superiors; R = to require feelings from someone to someone else. Then, it seems suitable to analyze the relationship as following a slightly different element layout: $R(x, y) \rightarrow \mathrm{R}(z, x)$.

[11] Both examples occur in *The New Rhetoric* (cf. Perelman and Olbrechts-Tyteca 1969, p. 222) and *The realm of rhetoric* (cf. Perelman 1982, p. 68) as illustrations of arguments utilizing similarity with logical principle of symmetry.

[12] It is obvious that logical implication (→) should be interpreted in this case as normative shift: e.g. 'should arise the commitment' etc.

It is obvious that identifying the second fragment as similar to logical symmetry (rather than not) by an audience depends on its factual scope of perceptiveness, tolerance and willingness to overlook possible important structural differences.

(b) A related problem lies in the fact that quasi-logical arguments differ from formal demonstrations not only at the level of syntax but also at the level of semantics. Quasi-logical arguments are arguments formulated in natural language, which include particular beings in specific semantic relations in the positions of variables and formal relations, for which some features of formal demonstrations are not binding. The elements of formal demonstrations can simply be filled with different semantic content in quasi-logical arguments. The decision on the permissible range of these possible interpretations, i.e., which types of entities and semantic relations are perceived as relevant instances of variables and relationships in a given formal demonstration, is again transferred to the audience's preferences. Let's illustrate it again with a few examples[13]:

(3) (…) he who has been generous in his wealth and merciful in his power has the right, so it seems, to appeal for generosity and mercy when fortune is unfavorable to him.
(4) If it is no disgrace for you to sell them, it is no disgrace for us to buy them.
(5) A mere trifle consoles us, for a mere trifle distresses us.

In all cases, we could agree that some mutual relations between individuals are implemented in all fragments. However, the type of mutuality is totally different: If we focus on analyzing the relationship R between the elements, it is obvious that only case (3) implements the same relationship between individuals, i.e. R = to be general and merciful. In cases (4) and (5), it seems more accurate to capture the relationships between individuals as pairs of relationships (R, R') among which a different semantic relationship is implemented.

In case (4), the relationship sell (R): buy (R') can be described from a semantic point of view as a pair of *indirect converses*: one act is viewed and described from two perspectives (cf. Cruse 1986, p. 234). However, in case (5), the relationship console (R): distress (R') seems more accurate to capture as a pair of *restitutives*: pairs, one of whose members necessarily denotes the restitution of a former state (e.g., damage: repair, remove: replace, stop: resume, kill: resurrect) (cf. Cruse 1986, p. 228).

The question is, how broad a scope in semantic interpretations of the relationship between R/R' is permitted by a given audience? Based on its cultural context, shared linguistic intuitions or also individual preferences, audiences can be more or less benevolent in their understanding mutuality: some audiences may accept only the first interpretation, some can also accept a second or even third semantic interpretation as cases of "symmetrical" relationships.

[13] All examples occur in *The New Rhetoric* (cf. 1969, pp. 221, 222, 224). Example (4) is also utilized by Perelman in *The realm of rhetoric* (cf. 1982, p. 68).

4.2 Practical Concept of Similarity

The theoretical concept of quasi-logical arguments presented in §45 of *The New Rhetoric* differs to a great extent from the presentation of individual quasi-logical argument sub-schemes in §46–59 of *The New Rhetoric* based on an empirical analysis of argumentative discourse.[14] In these paragraphs, Perelman and Olbrechts-Tyteca present ten of the most typical quasi-logical argument schemes, according to them. The ten schemes are distinguished on the basis of their similarity with a particular logical and mathematical principle. The basis of the typology is therefore the structural assimilation of arguments to formal demonstration, lexical assimilation is overshadowed in the typology. The particular, formal principles from the similarity to which we can, according to the authors, draw persuasiveness are captured in Table 1.[15]

Although Perelman and Olbrechts-Tyteca present quasi-logical arguments as imitating formal demonstrations, we can infer from the typology that they are not necessarily arguments formulated as a series of proof steps deriving the proposition from a set of initial assumptions. The arguments[16] fall into the category of quasi-logical arguments if their elements are at least in part in the structural layout of one of the above-mentioned formal principles (cf. Perelman 1979, p. 19). In practical concept, Perelman and Olbrechts-Tyteca offer a limited image of quasi-logical arguments: arguments utilizing a similarity with the structural layout of several well-known principles of propositional and predicate logic, arithmetic, or probability calculus.

The different character of the concept in the empirical part of *The New Rhetoric* can be linked with different starting points. The practical concept is not directly related to theoretical considerations based on the new rhetoric's program starting points, but results from the methodology of their empirical research. The argument schemes' typology was preceded by 10 years of collecting data from various fragments of argumentative discourses in which the values and actions are justified:

[14] In 1982, in *The realm of rhetoric*, Perelman introduces the typology of argument schemes including quasi-logical arguments. This typology is the shortened version presented in *The New Rhetoric*: he presents identical quasi-logical argument types, but rather limited descriptions of them and a smaller number of examples. Some rather limited descriptions of particular quasi-logical schemes can also be found in Perelman's *The New Rhetoric and the Humanities* (cf. 1979, pp. 19–21). For this reason, I refer primarily to the original typology presented in *The New Rhetoric* (1969).

[15] In order to identify the formal principles, I used primarily the authors' reference to these principles in the introductory part of chapters of *The New Rhetoric*, if they were available—as was the case in §50, 51, 53, 54, 56. I also took their formulations in *The realm of rhetoric* (Perelman 1982, Ch. 7) into consideration. If these references were not directly available or it was not sufficiently obvious which principle is referenced, I also investigated the specific explanations for individual variants and the structure of individual examples (§46, 55, 57, 58, 59).

[16] It is notable that Perelman and Olbrechts-Tyteca understand as an implementation of the quasi-logical scheme even cases in which only a premise/thesis is in the particular layout resembling a formal principle. For instance, authors present as a case of arguments of reciprocity the following statement, which they present as a Judeo-Christian ethical maxim (cf. 1969, p. 222): "Therefore all things whatsoever ye would that men should do to you, do ye even so to them." The proposition is not supported by a reason in their illustration and it seems appropriate to interpret it as an attempt to submit a standpoint, which the arguer commits to support in the case of doubt or disagreement.

Table 1 Typology of quasi-logical arguments and formal principles they imitate (based on Perelman and Olbrechts-Tyteca 1969, p. 195–255, cf. Perelman 1982, Ch. 7)

	Subtype of quasi-logical arguments	Imitated formal principle
Quasi-logical arguments	Contradiction and incompatibility (§46)	(a) Principle of contradiction
		Generally: two propositions *a* and non-*a* are mutually exclusive. In the new rhetoric, however, this principle is used in a more complex form: if *a* implies non-*c* and *b* implies *c*, then conjunction *a* and *b* implies *c* and non-*c*. Therefore, *a* and *b* are mutually exclusive
		(b) Principle of excluded middle
		Generally: Either the proposition or its negation is true, no third possibility is given. In the new rhetoric, it is used in a more complex form: if *a* implies non-*c* and *b* implies *c*, then conjunction *a* and *b* implies *c* and non-*c* and we can conclude that *a* and *b* are mutually exclusive. From mutual exclusivity follows that either *a* is true or *b* is true
	Identity and definition in argumentation (§50)	(a) Identity
		For all *x*, *x* is identical with *x*
		(b) Negation of principle of identity
		If *x* is not identical with *y*, then *x* has the property *A* and *y* does not, or it is not true that *x* and *y* have the property *A*
	The rule of justice (§52)	Scheme of rule for identity
		If *x* is identical with *y*, then *x* has the property *A* if and only if *y* has property A^{a}
	Arguments of reciprocity (§53)	Symmetry relation
		Relationship *R* is symmetric if and only if, whenever it is true that there is relationship *R* between *x* and *y*, it is true that there is relationship *R* between *y* and *x*
	Arguments of transitivity (§54)	Transitivity relation
		Relationship *R* is transitive if and only if whenever is true that there is relationship *R* between *x* and *y* and relationship *R* between *y* and *z*, it is true that there is relationship *R* between *x* and *z*

Table 1 (continued)

	Subtype of quasi-logical arguments	Imitated formal principle
Quasi-mathematical arguments	Inclusion of the part in the whole (§55)	(a) Whole-part axiom: The whole is greater than the part (b) Inclusion If set A is included in set B, then for every element x that belongs to A is true that x belongs to B
	Division of the whole into its parts (§56)	(a) Logical dilemma One (or both) of x or y is known to be true, but they both imply z, so regardless of the truth values of x and y we can conclude z (b) Exclusive disjunction The compound proposition x OR y is true if and only if one is true and the other is false
	Arguments by comparison (§57)	Transitivity of inequality If a is bigger than b and b is bigger than c, then is true that a is bigger than b and c
	Argumentation by sacrifice (§58)	Linear function Linear function is defined as having one independent variable x and one dependent variable y; a and b are constants: $y = ax + b$. Increase of x leads to increasing y
	Probabilities (§59)	Probability of phenomenon The probability P of a phenomenon A is calculated as a fraction of the number of favorable outcomes divided by the number of all possible outcomes

[a]In the case of the rule of justice, as well as in arguments of reciprocity and arguments of sacrifice, the reference to the formulation of an imitated formal principle is reductive in some way. For Perelman and Olbrechts-Tyteca, *normativity* is an important feature of the scheme. For example, the expression of a quasi-logical scheme (not identical with its formal counterpart) of rule of justice could be "If x and y belong to the same category A, x and y *should be* evaluated alike/treated alike" etc.

political, legal, philosophical, journalistic, etc. (cf. Perelman and Olbrechts-Tyteca 1969, p. 10). Although the authors do not offer a detailed explanation of their methodology, we can probably describe it as approaching the methodology of the empirically grounded theories (cf. Glaser and Strauss 1967; Strauss and Corbin 1998). Perelman and Olbrechts-Tyteca derive categories inductively, starting with data collection, i.e., textual fragments of arguments, after which they compare data specimens with each other and identify some prominent, repetitive features in the sample based on which they further generalize and construct more general categories. In this way, they create a hierarchy of categories, which is subsequently presented as a typology of argument schemes.

Perelman and Olbrechts-Tyteca offered a typology of argument schemes which is formulated based on their own reception of a large sample of argumentative fragments at the first level of analysis. Note that the authors were certainly not the intended addressees of these arguments. Nor do they seem to take into account the probable sums of the original addressees' interpretative choices in the analysis. Their analysis, therefore, does not respect the specifics of the original addressees: it is the discernment of schemes made by one particular audience, which accesses arguments in the sample as if it was their addressee. However, let us keep in mind that their own specific perception of arguments and the discernment of schemes does not need to be identical with the perceptions of those audiences who, on the contrary, were the real addressees of the arguments in the sample.

The fact that the types of relationship discerned and identified by Perelman and Olbrechts-Tyteca as the implementation of particular quasi-logical schemes often repeated in the sample serves as the reason for the authors claiming that (different) audiences are generally able to discern these schemes and generally acknowledge them as a non-problematic connection of elements. Such a claim is, however, sustainable only if we admit that all audiences perceive argument schemes in a similar way, i.e., they have roughly equal sums of interpretative choices, like Perelman and Olbrechts-Tyteca in the discernment of argument schemes. Regarding the reception of the argument schemes and their discernment, all audiences are in this sense unified by Perelman and Olbrechts-Tyteca as one audience, which corresponds to their own receptive abilities.[17] The concept of pluralist audiences proposed by the authors in a theoretical concept, is thus de facto replaced by the concept of one audience in the practical concept[18]:

> [Similarity$_2$] The argument implements a quasi-logical argument scheme if the connection of elements in an argument is *discerned* as similar to a formal principle by Perelman and Olbrechts-Tyteca.

[17] Perelman and Olbrechts-Tyteca can be considered in this respect as a specific audience, thanks to their education in logic and statistics (cf. Frank and Bolduc 2011, pp. 144–146). We can expect their higher sensitivity to structures that resemble formal principles. In other words, they can "find" the quasi-logical scheme even when no one else has identified it.

[18] The plurality of audiences is only guaranteed in comparison to the theoretical concept with regard to the accepted starting points, not with regard to the reception of argument schemes.

Let us illustrate. Specifically, the argument scheme of reciprocity is not presented as implemented at any time if it is discerned as such by an audience, but as the seven types of mutual relationships that Perelman and Olbrechts-Tyteca identified during their data collection and analysis as often repetitive. Note that the authors are criticized for vaguely describing these relationships (Table 2).

The definition appealing to the similarity perceived by the (different) audiences proposed by the theoretical concept is translated, in practice, into an enumerative definition in which specific interpretative choices of one particular audience are listed. The definition of similarity associated with the practical concept of quasi-logical arguments can be alternatively formulated regarding this enumerative strategy:

> [Similarity$_{2'}$] The argument implements a quasi-logical argument scheme if the connection of elements in the argument is the instance of specific semantic and syntactical relationships described by Perelman and Olbrechts-Tyteca in *The New Rhetoric* as an imitation of one of the ten well-known formal principles.

4.3 Interpreted Concept(s) of Similarity

The third concept that can be considered in connection with quasi-logical arguments can be found in the works of researchers who utilize Perelman and Olbrechts-Tyteca's typology of schemes in their own argumentative analyses or draw inspiration from it for their own theoretical purposes. Typically, these researchers explicitly refer to the ten schemes formulated in *The New Rhetoric* and—aware of the vagueness of the scheme descriptions listed therein—offer explicates of the schemes based on their own interpretative choices and formed by the terms of their target theoretical frames. Thus, we can say that the interpreted concepts are sets of specific explicates of schemes offered by Perelman and Olbrechts-Tyteca in their practical concept. Depending on how many interpreters we have, we can get as many specific interpreted quasi-logical argument concepts.

The variability in the interpretations of quasi-logical arguments' practical concept by various interpreters can again be illustrated in the arguments of reciprocity originally considered by Perelman and Olbrechts-Tyteca as imitating a logical relationship of symmetry. The scheme of arguments of reciprocity explicitly adopted from Perelman and Olbrechts-Tyteca's typology can, for instance, be found in the works of Garssen (1997 according to van Eemeren et al. 2007, p. 139), Warnick (2000), or Grasso (2003, cf. 2002). Although the authors interpret the same paragraph of *The New Rhetoric*, it is indicative that they come up with different explicates. The final expression of the scheme is influenced not only by their own perception, but also by the target theoretical frame of explication.

Garssen is embedded in the pragma-dialectical approach to argumentation, he interprets the argument of reciprocity as a subtype of analogical argumentation (cf. van Eemeren et al. 2007, p. 139). In his approach, the presence of specific principle of reciprocity ("giving in return") as an argument principle connecting reason with

Table 2 Practical concept of arguments of reciprocity: variants discerned by Perelman and Olbrechts-Tyteca (cf. 1969, pp. 221–225, cf. Perelman 1982, p. 67–70, 1979, p. 21)

Name of variant, description	Illustration
Antecedent-consequent These arguments by reciprocity, based on the connections between the antecedent and the consequent of the same relation, appear, more than any other quasi-logical arguments, to be at once formal and based on the nature of things. Symmetry is most often assumed in the very qualification of situations.	If it is no disgrace for you to sell them, it is no disgrace for us to buy them
Transposition of points of view Arguments based on reciprocity can also result from the transposition of points of view, which makes it possible through their symmetry to recognize the identity of certain situations	He who has been generous in his wealth and merciful in his power has the right, so it seems, to appeal for generosity and mercy when fortune is unfavorable to him
The same effect Two actions which are complementary, in the sense that they both constitute a necessary condition for bringing about a certain effect, can provide an occasion for using the argument from symmetry	"I have accused; you have condemned," is the famous reply of Domitius Afer
Complementarity Two actions which are complementary, in the sense that they both constitute a necessary condition for bringing about a certain effect, can provide an occasion for using the argument from symmetry	"If you grant this authorization and the King of Holland refuses, what moral prestige will result for you! If you refuse the authorization and the king of Holland grants it, what prestige will accrue to him!
Inversion Quasi-logical arguments can make use of another type of symmetry, resulting from the fact that two actions, two courses of behavior, two events, are presented as the inverse of each other. It is concluded that what applies to one—means necessary to achieve it, evaluation of it, nature of the event—likewise applies to the other	A mere trifle consoles us, for a mere trifle distresses us
Contraries Most of the examples of argument by contraries given to us by the ancients end up with a generalization, starting from some particular situation and requiring that the same treatment be applied to the symmetrical situation	How can it be maintained that on sufficient proof the judge ought to give a verdict against an innocent man whose innocence happens to be known to him, and that in the absence of sufficient proof he ought to absolve the guilty, even though he happens to have knowledge of his crime
Generalization Use of the argument by reciprocity is the basis of a generalization frequently encountered in philosophy, such as that which asserts that all that is born dies, thus going from the birth of a being to his contingency	All that is born dies

standpoint is crucial for identifying the scheme.[19] On the contrary, Warnick emphasizes in formulating a scheme that the arguments are always situated and contextualized. She interprets the scheme in cultural, social and contextual terms (cf. Warnick 2000, p. 108) and offers variable examples of arguments of reciprocity utilizing different types of mutuality (cf. 2000, 110–111).

Grasso develops in her approach a model of rhetorical argumentation, she takes inspiration from theories on discourse structure, and also emphasizes the role of discourse and language (Grasso 2003, p. 25). According to her, a rhetorical scheme "is meant to express when it is admissible to use a given relationship, by capturing the restrictions that, from case to case, have to apply for the relationship to be used legally, or safely." (Grasso 2002, p. 3) She conceptualizes four sub-schemes of reciprocity in terms of different ontological, acceptability, relevance and sufficiency constraints (cf. Grasso 2003, pp. 180–181).

Considering these target frames into which Perelman's original scheme is transferred, it is clear that the resulting explicates reflect different theoretical purposes and may have differing scopes of application. The comparison presented in Table 3 appropriately illustrates the variability of interpretations of what reciprocity is according to authors.

The similarity of arguments with formal demonstrations is conceived, as in the case of the practical concept, as dependent on a particular addressee's receptive abilities—in this case, on a particular interpreter of a category described in *The New Rhetoric*:

> [Similarity$_3$] The argument implements a quasi-logical argument scheme if the connection of elements in an argument is *discerned* by the interpreter as similar to the formal principle captured in *The New Rhetoric*.

The fact that there may be different interpreted quasi-logical argument concepts based on a different conceptual basis leads to the need to carefully distinguish between them. As can be seen from the comparison, categorization tools marked by the same term may not be identical and can refer to completely different argumentation technique types. Given different theoretical basis they also can function in different way in research. Thus, the results of applying such tools in the analysis do not necessarily overlap with Perelman and Olbrechts-Tyteca's results, nor with each other (cf. van Eemeren et al. 1996, p. 123). In this sense, it seems to be safer to index interpreted scheme concepts. For example, to the interpreted arguments of reciprocity concepts, it would be referred to as "Garssen's (pragma-dialectical)", "Grasso's", or "Warnick's" quasi-logical reciprocal scheme, etc.

[19] In pragma-dialectics, argument schemes are defined by unique argument principles and specific sum of critical questions. Argument of reciprocity is defined by the reciprocal argument principle. According to pragma-dialecticians, "This principle particularly applies to people (and countries), and implies that one person should act towards another person in the same way the other person act towards him." (van Eemeren et al. 2007, p. 139). Three critical questions are connected with the scheme: Should people in the same situation be treated equally in this case? Do the compared things show sufficient similarities to justify the proposed equal treatment? Are there any relevant differences between the compared things that stand in the way? (van Eemeren et al. 2007, p. 139).

Table 3 Interpreted concepts of arguments of reciprocity

Garssen 1997 (according to van Eemeren et al. 2007, p. 139)	Warnick (2000, pp. 121–122)	Grasso (2003, pp. 180–181)[a]
Reciprocity as "Giving in return" Ex.: *You should help me, because I helped you before*	Reciprocity as two beings or situations treated equivalently[b] Ex.: *And if I did not publish the required number of articles, then I would not be promoted*	Reciprocity as a symmetrical property shared by two situations Ex.: *Since you like biscuits, you should like apples too, as apples are as crunchy as biscuits* "Complementarity" as two situations are both a necessary condition to bring about a certain effect Ex.: *You cannot concentrate on eating less alone, without also making some exercise* "Contraries" as two states of affairs, one the opposite of the other Ex.: *If you like to eat, you should like to fast* "Inversion" as two actions, one the inverse of the other Ex.: *If you like to eat, you should like to cook*

The used examples are original, taken from the works of the authors. An exception is the example of Garssen's concept, which was constructed based on the presented definition (cf. van Eemeren et al. 2007, p. 139)

[a]In the table, a simplified version of Grasso's schemes is presented, for the purposes of comparison only. A complete description of schemes (included their description in terms of various types of constraints) is presented in Grasso (2003, pp. 180–181)

[b]Warnick's complete description of a scheme is: "Structurally similar to the conditional syllogism. Two beings or situations treated equivalently. Both are reduced so as to make them appear symmetrical." (2000, pp. 121–122)

Table 4 Three concepts of quasi-logical arguments

	Theoretical concept	Practical concept	Interpreted concept(s)
Source of concept	Theoretical passages of *The New Rhetoric*, §45	Presentation of schemes in *The New Rhetoric*, §46–56; *The realm of rhetoric*, Ch. 7	Works of researchers applying concepts of the new rhetoric in analyses or inspired by them
Concept of similarity	[Similarity$_1$] The argument implements a quasi-logical argument scheme when the connection of elements in an argument is discerned by an audience as similar to a particular formal demonstration	[Similarity$_2$] The argument implements a quasi-logical argument scheme if the connection of elements in an argument is discerned by Perelman and Olbrechts-Tyteca as similar to a formal principle	[Similarity$_3$] The argument implements a quasi-logical argument scheme if the connection of elements in an argument is discerned by the interpreter as similar to the formal principle captured in *The New Rhetoric*
Starting point of the concept of similarity	It is based on the concept of argumentation as aimed on audiences with specific receptive abilities. The implementation of the quasi-logical scheme can be perceived by one audience and not by others	It is based on Perelman and Olbrechts-Tyteca's analysis of empirical data. The implementation of the quasi-logical scheme is assessed with respect to the receptive abilities of a particular audience: Perelman and Olbrechts-Tyteca It is (one) specific case of theoretical concept	It is based on an attempt of (different) interpreters to provide clear definitions of ten argument schemes provided by Perelman and Olbrechts-Tyteca in *The New Rhetoric* in terms of target theoretical frame into which the original concept is transferred They are specific explicates of the practical concept

4.4 Three Different Concepts

Let us conclude that three quasi-logical argument concepts can be distinguished given the source where they are located, the initial concept of similarity and the starting point for such a conceived similarity. Differences in concepts are captured in Table 4.

In the light of this analysis, one of the sources of the current failure and non-attractiveness of quasi-logical arguments for modern argumentation theorists can lie in the confusion of the concepts under one shared term. The comparison also reveals the practical concept as having a *privileged position* in the theory compared to the theoretical and interpreted concepts, because it precedes them theoretically and in terms of time. The theoretical concept arises from extending the validity of Perelman and Olbrechts-Tyteca's conclusions about the particular group of arguments' effectiveness for one audience (achieved in empirical research) to other potential audiences. The motif for this use of the original practical concept is their attempt to offer a general theory of argumentation based on the idea of pluralistic audiences. Thus, we can say that practical concept precedes theoretical concept and enables its formulation. Interpreted concepts, on the contrary, arise from attempts to offer unambiguous explanations for the practical concept. Therefore, the practical concept precedes the emergence of individual interpreted concepts too.

5 Effectiveness of Arguments for the Audience

After investigating similarity, we can focus on examining the second component of the thesis: claim of effectiveness of arguments that are similar to formal demonstrations. What should—according to the authors—fulfill an argument in order to be effective regarding the intended audience? We can proceed from the general definition of effective arguments that the authors put forward in *The New Rhetoric*:

> An efficacious argument is one which succeeds in increasing this intensity of adherence among those who hear it in such a way as to set in motion the intended action (a positive action or an abstention from action) or at least in creating in the hearers a willingness to act which will appear at the right moment. (Perelman and Olbrechts-Tyteca 1969, p. 45)

The effectiveness of an argument is defined by the author as a property we can attribute to the argument depending on the audience's reception of the argument, i.e., it is also an *audience-dependent* property. An effective argument is generally an argument whose submission raises a particular response by an audience: it leads to/increases the tendency of the audience to accept the presented standpoint. In view of the new rhetoric emphasizing practical reasoning, the effectiveness of the argument is considered in conjunction with the addressee's willingness to take action in accordance with such a standpoint.

In order to lead to an audience's acceptance, Perelman and Olbrechts-Tyteca assume two necessary conditions: the argument must be based on the *starting points* adhered by the audience and must implement an *argument scheme* acknowledged by the audience:

> Our analysis of argumentation will deal first with what is taken as the starting point of arguments and afterwards with the way in which arguments are developed through a whole set of associative and dis-associative processes. This division of the subject is indispensable to our examination and should not be misunderstood. *The unfolding* [i.e. argument scheme] *as well as the starting point of the argumentation presuppose indeed the agreement of the audience*. This agreement is sometimes on explicit premises, sometimes on the particular connecting links used in the argument or on the manner of using these links: from start to finish, *analysis of argumentation is concerned with what is supposed to be accepted by the hearers*. (Perelman and Olbrechts-Tyteca 1969, p. 65, italics added, I. S.)

Two general conditions for the effectiveness of arguments according to the new rhetoric are captured in Scheme 1.

After introducing the general concept of an argument's effectiveness from the perspective of the new rhetoric, we can properly grasp the way in which *argument schemes* particularly can contribute to the effectiveness of an argument. Firstly, the argument scheme is only one of the two necessary conditions for an argument's effectiveness. The argument scheme contributes to the standpoint's acceptance only if the argument is also based on the starting points accepted by the audience.

Secondly, in order to contribute to the standpoint's acceptance, it is necessary for the scheme to be *acknowledged* by the audience. The scheme can only contribute to the acceptance of the standpoint if the audience considers it to be a relevant, usual, non-problematic way of connecting or separating the elements in the argument (cf. Perelman and Olbrechts-Tyteca 1951, pp. 252–253). The usual ways of doing this are included between the elements agreed by the audience. According to Perelman and Olbrechts-Tyteca, audiences accept not only facts or values (i.e. starting points), but also "constructs of reality and connections between facts and values—in short, a totality of common beliefs, which shall call "places", in recognition of the ancient usage of the term "commonplaces". (Perelman and Olbrechts-Tyteca 1951, p. 252) From here, we can paraphrase the new rhetoric's *general conception of the effectiveness* of the argument scheme as follows:

> [Effectiveness] The argument scheme contributes to the acceptability of the standpoint if it is *acknowledged* by the audience as a relevant way of connection/separation of elements in an argument.

5.1 Effectiveness of Quasi-logical Arguments

The thesis of effectiveness of quasi-logical arguments connects the argument's effectiveness with their similarity to formal demonstrations: quasi-logical arguments are *effective because of their similarity* to formal principles. According to Perelman and Olbrechts-Tyteca, their persuasive strength is *derived* from their similarity to formal principles (cf. 1969, p. 193).[20]

[20] The claim of effectiveness is not associated with any particular concept of similarity. Therefore, we can deal with the effect on the audience/audiences in general, independently on this distinction.

Analyzing the concepts of similarity and effectiveness in relation to the audience implemented in previous sections has led us to interpret both terms as *audience-dependent* properties of argument. In order for the argument to be effective based on a similarity with a formal principle, two types of reception on the audience's side must take place: the *discernment* of quasi-logical character of the connection of elements in the argument and the *acknowledgment* of this connection as relevant, usual and unproblematic. This general idea of quasi-logical argument effectiveness can be captured by Scheme 2.

Note that the thesis of the effectiveness of quasi-logical arguments as formulated above can be specifically captured as a relationship between fields (a) and (b) in Scheme 2, and only concerns the *level of the argument scheme*: (b) connecting the elements in the argument is *acknowledged* by the audience as relevant because (a) it is *discerned* by the audience as similar to a formal principle.

The important question is, what is the character of the relationship between these two types of reception of audience? Why would people tend to acknowledge an argument that they discern (more or less consciously) as having quasi-logical character? Based on the works of Perelman and Olbrechts-Tyteca, we can consider two different perspectives on effective quasi-logical arguments: in terms of socially shared standards of community or in terms of individual argument processing.

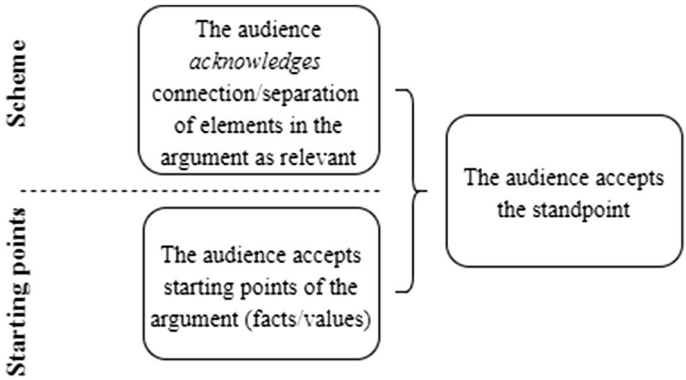
5.1.1 Quasi-logical Arguments as "Logic of Common Folk"

In some passages of his later texts, Perelman considers that using quasi-logical arguments *precedes* familiarity of audience with formal logic or mathematics principles (cf. Perelman 1979, 1982). It can be said (in terms of our analysis) that acknowledging quasi-logical element connection in arguments as unproblematic seems *independent* of the audience's acquaintance with the formal principles. Particularly, Perelman admits that people in antiquity used quasi-logical arguments, although in principle they could not know their formalized counterparts:

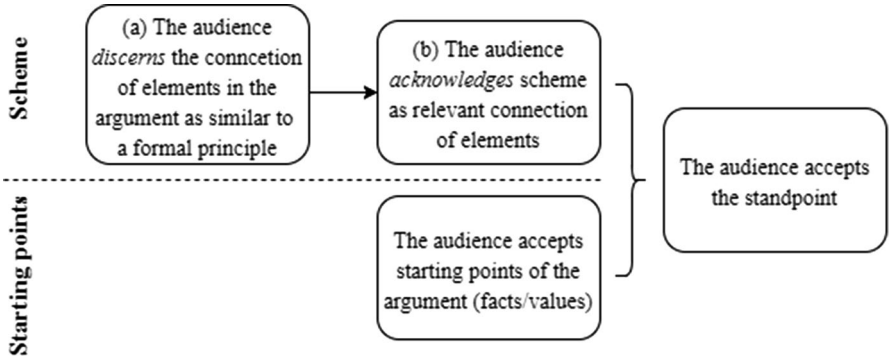
> Let us note in passing that in antiquity, when the mathematical version of scientific thinking was less developed, recourse to quasi-logical arguments was more frequent. (Perelman 1982, p. 53)

> These [quasi-logical] arguments are similar to the formal structures of logic and mathematics. In fact, *men apparently first came to an understanding of purely formal proof by submitting quasi-logical arguments*, such as many of the loci listed in Aristotle's Topics, *to an analysis that yielded precision and formalization*. (Perelman 1979, p. 19, italics added, I.S.)

We can consider in directing this interpretation that quasi-logical element connection is conceived by the audience as a usual, unproblematic way of connecting the elements, because this type of connection was historically established in antique argumentative practice, or more generally shared and transmitted in



Scheme 1 General conditions of effectiveness of an argument according *The New Rhetoric*



Scheme 2 Effectiveness of arguments implementing quasi-logical argument scheme

western culture and between members of this culture.[21] As Warnick and Kline speculate at this point, the success of quasi-logical connecting elements in our (western) culture could be explained by reference *to this culture's general preferences*: "Contemporary culture's preference for certainty, simplicity, and parsimony contributes to the effectiveness of quasi-logical arguments." (Warnick and Kline 1992, p. 6)

In this sense, we can conceive an acknowledged sum of schemes as *socio-logic*—they can be interpreted as part of a „socially developed sense of practical reasoning." (Goodwin and Wenzel 1979, p. 289; cf. Leff 1983, p. 25; Warnick 2000, p. 108). Goodwin and Wenzel consider our ability to discern and acknowledge justifying schemes as socially formed, confirmed, and transmitted. Similarly, Warnick considers that inference patterns are recursive and recognizable in the culture in which

[21] Comparison of typologies of schemes (*loci*) distinguished by Aristotle and Perelman and Olbrechts-Tyteca implemented by Warnick (cf. 2000) shows that some quasi-logical schemes do not, in fact, occur in antiquity (e.g. probability calculus—in form of statistical and numerical probability) and on the contrary, often occur in contemporary culture.

they are used; they may or may not occur in other cultural contexts. "But once a topical pattern has developed into common use, it will be used over and over in various manifestations and will be effective by virtue of its recognizability." (Warnick 2000, p. 110) In directing this interpretation, people can tend to accept quasi-logical arguments because they are members of a community in which this type of formulating and evaluating argumentation is usual. The explanation of effective quasi-logical arguments can, from this perspective, be formulated as follows:

> [Effectiveness$_1$] Discernment of the quasi-logical connection of elements in the argument leads to the unproblematic acknowledgment of the argument by the audience based on the fact the quasi-logical element connections in the argument are part of the socially shared standards for formulating and evaluating argumentation.

However, an important consequence of such an interpretation is that quasi-logical arguments lead to acceptance by the audience *not directly on the basis of their similarity* to formal principles but because *they belong to the sum of the usual schemes of reasoning*. In principle, it is possible that quasi-logically formulated arguments will be acknowledged by an addressee who does not know and has never been acquainted with the principles of formal logic or mathematics. It is crucial for the addressee to adhere a quasi-logical principle accepted by the community, not to know its logical or mathematical counterpart. For instance: it seems crucial for the addressee to acknowledge the reciprocal linking of elements in argument as drawing on adhered values or preferences of her cultural frame, e.g. its specific understanding and evaluating mutuality, not necessarily acquainted with the logical principle of symmetry.

The similarity of quasi-logical principles with formal demonstrations is "subsequent" in this sense. It does not serve as a description of a source of effective arguments for audience. It is rather a label introduced by the analyst for a specific set of types of argumentative connections. The label thus chosen allows other analysts—familiar with principles of logic or mathematics—to easily identify certain types of argument structure in informal texts. Interpreting similarity as only an analytical label, rather than a description of the source of effectiveness can be supported by the wording that Perelman uses in introducing the category:

> Quasi-logical arguments are those which *can be understood by way of comparison* to logical, mathematical, formal thinking. (Perelman 1982, p. 50, italics added, I.S.)

> *We will classify* quasi-logical arguments *by comparing them to formal reasoning*, with which they have something in common; then we will have to draw attention to the things that distinguish them from formal reasoning, that is, to the thing which give rise to controversy and thereby make them nonrigorous (Perelman 1982, p. 53, italics added, I.S).

5.1.2 Quasi-logical Arguments as Imitations of Successful Formalized Logic/ Mathematics

However, other comments provided by the authors may lead us to a different perspective, according to which the acknowledgment of quasi-logical arguments is *directly dependent* on an audience's acquaintance with the formal principles. From this perspective, the notion of similarity is not only an analytical label but also primarily a description of the source of effectiveness. Perelman and Olbrechts-Tyteca mention that quasi-logical arguments derive their persuasive strength from similarity, or that the speaker can refer explicitly to those formal structures from which the similarity is drawn and utilize it:

> But, since there are formal proofs of recognized validity, *quasi-logical arguments derive their persuasive strength from their similarity with these well-established modes of reasoning.* (Perelman and Olbrechts-Tyteca 1969, p. 193, italics added, I.S.)

> Quasi-logical arguments are presented in a more or less explicit manner. Sometimes the *speaker will mention the formal reasoning to which he makes reference, availing himself of the prestige of logical thought;* sometimes the formal reasoning provides only an underlying texture. (Perelman and Olbrechts-Tyteca 1969, p. 194, italics added, I.S.)

Based on these comments, it seems that the quasi-logical connection of elements draws its acknowledgment from the fact that there are formal principles recognized by the audience as generally valid, and these formal principles are somehow associated with a quasi-logical scheme in the audience's perception.

> [Effectiveness$_2$] Discernment of the quasi-logical connection of elements in an argument leads to an unproblematic acknowledgment of the argument by the audience, based on the fact that the quasi-logical element connections in the argument evoke the principles of formal logic or mathematics that the addressee recognizes as generally valid.

In this interpretation, quasi-logical arguments are *imitations* of formal principles. They utilize the possibility that discerning quasi-logical character can evoke addressee the formal principle accepted by the addressee as valid in the domain of logic or mathematics.[22] In this interpretation, however, an effect of quasi-logical arguments assumes an audience's preceding acquaintance with the principles of formal logic or mathematics. Quasi-logical element connection in arguments would not be acknowledged by an audience that is not—at least at an elementary level—familiar with the formal principles of logic or mathematics and their assumed credibility.

This course of interpretation is mainly followed by Perelman's interpreters. For example, Dearin conceives "prestige associated with rigorous thought" as

[22] Authors explicitly admit a prestige of logical thinking in the following passage: "Doubtless, logical proof can itself be the object of the hearer's attention: he admires its elegance, deplores its heaviness, notices its adequacy to the end that is pursued." (Perelman and Olbrechts-Tyteca 1969, p. 190).

"accumulation of mental associations an audience in our culture might hold as a result of the generally assumed credibility of modern logic, or the unchallenged success of mathematical reasoning in certain well-delineated areas of knowledge." (Dearin 1982, p. 85)

Van Eemeren is also inclined towards this interpretation. He develops an explanation of the relation conditions between discerning the scheme and its effect on the audience by focusing on the speaker's role—it is her task *to ensure* that the audience *discerns the argument as imitating* the formal principle:

> The use of suitable terms and premises is not enough for quasi-logical argumentation to have the desired effect; it must also be ensured that the audience *recognizes* the form of the argumentation as logical. In order to emphasize the similarity with argument forms from mathematics or logic, speakers and writers will have to do their best to mark the form of their argumentation as clearly as possible. (van Eemeren et al. 1996, p. 108, italics in original, cf. 2014, p. 272)

The effect of quasi-logical argumentation, is according to him, based on the fact that logic and mathematics have gained considerable prestige, and "the arguer attempts to radiate the prestige of logic and mathematics to argumentation in ordinary language." (van Eemeren et al. 2014, p. 272)

5.1.3 Two Explanations of Effectiveness of Quasi-logical Arguments

As shown, based on Perelman and Olbrecht-Tyteca's comments, the effectiveness of quasi-logical arguments can be explained from two possible perspectives based on emphasizing one of the possible aspects of effectiveness: socially shared justifying standards of given culture and individual argument processing. Crucial features of both explanations are summarized in Table 5.

Due to the different conceptual starting points of both perspectives, both explanations are, in principle, compatible in an effort to explain the effect of quasi-logical arguments in contemporary western culture, where we can consider that addressees are likely familiar with the principles of formal logic or mathematics or with the ethos of their credibility. However, in cases where we cannot expect this context, i.e. in cases of argumentation their submission preceded the process of mathematical and logical formalization, it is not so. The explanation of effectiveness referring to individual processing via the association quasi-logical scheme with particular formal principle cannot be meaningfully maintained. It can be concluded that the concept of the effect of quasi-logical arguments' usefulness for future application in researching the effectiveness of argumentation depends on the investigation of both explanations and answering the question of which of them seems more adequate, or finding a way to adequately balance them.

Table 5 Two explanations of effectiveness of quasi-logical arguments

Source of interpretation	Perelman (1979, 1982)	Perelman and Olbrechts-Tyteca (1969) (§45)
Explanation of effectiveness of quasi-logical argument	[Effectiveness$_1$] Discernment of the quasi-logical connection of elements in the argument leads to the unproblematic acknowledgment of the argument by the audience based on the fact the quasi-logical element connections in the argument are part of the socially shared standards for formulating and evaluating argumentation	[Effectiveness$_2$] Discernment of the quasi-logical connection of elements in an argument leads to an unproblematic acknowledgment of the argument by the audience, based on the fact that the quasi-logical element connections in the argument evoke the principles of formal logic or mathematics that the addressee recognizes as generally valid
Starting points of explanation:	Emphasis on social (cultural) context enabling occurrence and success of quasi-logical argument Quasi-logical connection of elements in argument is conceived as generally acknowledged by the community preferring simplicity or parsimony	Emphasis on individual processing the quasi-logical argument by an addressee Acknowledgment of quasi-logical connection of elements in argument is conceived as association of quasi-logical principle with its formal counterpart in the mind of the addressee
Consequences of explanations	Explanation of effect of argument *is not dependent* of assuming the addressee's acquaintance with formal principles of logic or mathematics	Explanation of effect *is dependent* of assuming the addressee's acquaintance with formal principles of logic or mathematics

6 Conclusion

This study focused on reconstructing and analyzing the elements of the thesis of the effectiveness of quasi-logical arguments. In the first part, historical impulses shaping the form of new rhetoric were specified. The reasons that led Perelman and Olbrechts-Tyteca to the introduction of a sharp distinction between logic and argumentation were shown. It was shown that this sharp distinction leads to the postulation of the anthropological concept of reasonableness for the domain of argumentation.

In the second part, a general definition of quasi-logical arguments was presented, and the thesis of their effectiveness was formulated. Two constitutive components of the thesis were specified: a definition of arguments based on similarity with formal demonstrations and the claim to the effectiveness of these arguments. Both elements were analyzed and their definitions in relation to the audience were specified.

In the third part, it was shown that, based on the question in which audience is considered as the addressee of argumentation, we can distinguish three concepts of similarity in argument and therefore three concepts of quasi-logical arguments. The theoretical concept considers audience pluralism based on their individual reception of similarity. The practical concept attributes the similarity on the basis of its discernment by Perelman and Olbrechts-Tyteca. The interpreted concept attributes similarity with respect to its discernment by specific analysts. The result of this analysis was presented by specified definitions, sources and the bases of these concepts. It was shown that practical concept has a privileged position in theory because it precedes theoretical and interpreted concepts.

The analysis of the claim of effectiveness in the fourth part showed that the effectiveness of quasi-logical schemes can be understood as acknowledging the relevance of the quasi-logical element connection in the argument by the audience. It was showed that based on Perelman and Olbrechts-Tyteca's works, two possible interpretations of this relationship can be formulated, depending on whether we emphasize the social (cultural) or individual aspects of their explanation. It was shown that these interpretations draw on a different initial concept of the audience's relationship with the formal principles of logic or mathematics: in the first explanation, it is not expected, but in the second it is.

The analysis of two components of the thesis of effectiveness of quasi-logical arguments shows that the category of quasi-logical arguments as defined by the new rhetoric includes ambiguous constituents. A researcher who wants to use this category in her application should therefore clarify which concept of similarity she uses in identifying or reconstructing phenomena and, potentially, which perspective on effectiveness she prefers in her attempt to analyze or evaluate the effect of argument thus identified.

Acknowledgments I wish to thank Marek Picha, Jiří Raclavský and both anonymous reviewers for their helpful comments to earlier versions of this paper. I also wish to thank Bart Garssen, Tomáš Ondráček and Andreas Finsen for very helpful and inspiring discussions regarding quasi-logical arguments.

References

Aristotle. 1934. Nicomachean Ethics. In *Aristotle in 23 Volumes*, Vol. 19, translated by H. Rackham. Cambridge, MA: Harvard University Press; London: William Heinemann Ltd. Available at: http://www.perseus.tufts.edu/hopper/text?doc=urn:cts:greekLit:tlg0086.tlg010.perseus-eng1:1094b. Accessed 7 May 2018.

Bochenski, Józef M. 1981. The general sense and character of modern logic. In *Modern logic—A survey. Historical, philosophical and mathematical aspects of modern logic and its applications*, ed. Evandro Agazzi, 3–14. Dordrecht: Springer.

Cruse, D.Alan. 1986. *Lexical semantics*. Cambridge: Cambridge University Press.

Dearin, Ray D. 1982. Perelman's concept of "quasi-logical" argument. A critical elaboration. In *Advances in argumentation theory and research*, ed. J.Robert Cox and Charles A. Willard, 78–94. Carbondale: Southern Illinois University Press.

Frank, David A., and Michelle Bolduc. 2011. Lucie Olbrechts-Tyteca's New Rhetoric. *Quarterly Journal of Speech* 96 (2): 141–163.

Garssen, Bart. 1997. *Argumentatieschema's in PragmaDialectisch Perspectief. Een Theoretisch en Empirisch Onderzoek* (doctoral thesis). Dordrecht: ICG Printing b.v.

Glaser, Barney, and Anselm L. Strauss. 1967. *The discovery of grounded theory: Strategies for qualitative research*. New Brunswick, London: Aldine Transaction.

Goodwin, Paul D., and Joseph Wenzel. 1979. Proverbs and practical reasoning: a study in socio-logic. *Quarterly Journal of Speech* 65 (3): 289–302.

Grasso, Floriana. 2002. Towards a framework for rhetorical argumentation. In *Proceedings of the sixth workshop on the semantics and pragmatics of dialogue (EDILOG 2002), 4–6 September 2002, Edinburgh, UK*, Johan Jos, Mary E. Foster and Colin Matheson (Eds.), pp. 53–60. https://pdfs.semanticscholar.org/dbbb/3ac9daa04e7fe5a39495e3887e5de9d1d033.pdf. Accessed 7 May 2018.

Grasso, Floriana. 2003. *Characterising rhetorical argumentation* (doctoral thesis). http://cgi.csc.liv.ac.uk/~floriana/TesiFlo.pdf. Accessed 7 May 2018.

Gross, Alan G., and Ray D. Dearin. 2003. *Chaim Perelman*. Albany: State University of New York Press.

Haynes, David N. 1981. The language and logic of law: A case study. *University of Miami Law Review* 35 (2): 183–254.

Kienpointner, Manfred. 1993. The empirical relevance of Perelman's new rhetoric. *Argumentation* 7 (4): 419–437.

Kienpointner, Manfred. 2002. Perelman on causal arguments: The argument of waste. *ISSA Proceedings 2002*. Available at: http://rozenbergquarterly.com/issa-proceedings-2002-perelman-on-causal-arguments-the-argument-of-waste/. Accessed 7 May 2018.

Leff, Michael C. 1983. The topics of argumentative invention in Latin rhetorical theory from Cicero to Boethius. *Rhetorica: A Journal of the History of Rhetoric* 1 (1): 23–44.

Perelman, Chaïm. 1954. Proof in philosophy. *Hibbert Journal* 52 (4): 354–359.

Perelman, Chaïm. 1955. How do we apply reason to values? *The Journal of Philosophy* 52 (26): 797–802.

Perelman, Chaïm. 1979. *The new rhetoric and the humanities*. Dordrecht: Reidel Publishing Company.

Perelman, Chaïm. 1980. *Justice, law and argument: Essays on moral and legal reasoning*. Riedel, Dordrecht: D. Reidel Publishing Company.

Perelman, Chaïm. 1982. *The realm of rhetoric*. Notre Dame: University of Notre Dame Press.

Perelman, Chaïm. 1986. Old and new rhetoric. An address delivered by Chaim Perelman at Ohio State University, November 16, 1982. In *Practical reasoning in human affairs: Studies in honor of Chaim Perelman*, ed. James L. Golden and Joseph J. Pilotta, 1–18. Dordrecht: Springer.

Perelman, Chaïm. 1989. Formal logic and informal logic. In *From metaphysics to rhetoric*, ed. Michel Meyer, 9–14. Dordrecht: Kluwer Academic Publishers.

Perelman, Chaïm, and Lucie Olbrechts-Tyteca. 1951. Act and person in argument. *Ethics* 61 (4): 251–269.

Perelman, Chaïm, and Lucie Olbrechts-Tyteca. 1957. The new rhetoric. *Philosophy Today* 1 (1): 4–10.

Perelman, Chaïm, and Lucie Olbrechts-Tyteca. 1969. *The new rhetoric: A treatise on argumentation*. Notre Dame: University of Notre Dame Press.

Strauss, Anselm L., and Juliet M. Corbin. 1998. *Basics of qualitative research: Techniques and procedures for developing grounded theory*. Thousand Oaks: Sage Publications Inc.

Toulmin, Stephen E. 1976. *Knowing and acting. An invitation to philosophy. New York*. London: Macmillan.

van Eemeren, Frans H. 2009. Argumentation theory after the New Rhetoric. *L'analisi linguistica e letteraria* 17 (1): 119–148.

van Eemeren, Frans H., and Rob Grootendorst. 1988. Rationale for a pragma-dialectical perspective. *Argumentation* 2 (2): 271–291.

van Eemeren, Frans H., Rob Grootendorst, and A. Francisca Snoeck Henkemans. 1996. *Fundamentals of argumentation theory*. New York: Routledge.

van Eemeren, Frans H., Peter Houtlosser, and A. Francisca Snoeck Henkemans. 2007. *Argumentative indicators: A pragma-dialectical study*. Dordrecht: Springer.

van Eemeren, Frans H., Bart Garssen, Erik C.W. Krabbe, A. Francisca Snoeck Henkemans, Bart Verheij, and Jean H.M. Wagemans. 2014. *Handbook of argumentation theory*. Dordrecht: Springer.

van Rees, Agnes. 2008. *Dissociation in argumentative discussions. A pragma-dialectical perspective*. Berlin: Springer.

Walton, Douglas, Chris Reed, and Fabrizio Macagno. 2008. *Argumentation schemes*. Cambridge: Cambridge University Press.

Warnick, Barbara. 2000. Two systems of invention: The topics in the rhetoric and the new rhetoric. In *Rereading Aristotle's rhetoric*, ed. Alan G. Gross and Arthur E. Walzer, 107–129. Carbondale: Southern Illinois University Press.

Warnick, Barbara, and Susan L. Kline. 1992. The new rhetoric's argument schemes: A rhetorical view of practical reasoning. *Argumentation and Advocacy* 29 (1): 1–15.

Wenzel, Joseph. 1990. Three perspectives on argument: Rhetoric, dialectic, logic. In *Perspectives on argumentation: essays in honor of Wayne Brockriede*, ed. Wayne Brockriede, Robert Trapp, and Janice E. Schuetz, 9–26. Prospect Heights, IL: Waveland Press.

Wenzel, Joseph. 1992. Perspectives on argument. In *Readings in argumentation*, ed. William L. Benoit and Dale Hample, 121–144. Berlin: Walter de Gruyter.